

ECE 315: Signals and Systems

Midterm Exam I Sample

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This sample is for self-study in preparation for Midterm Exam I; solutions will not be released. The exam covers Chapters 1 and 2 of **SS**, as well as lecture notes and homework through **Feb 27**. A summary of key concepts and related sample questions is provided below.

- Basics of discrete- and continuous-time signals: discrete vs continuous, signal size, signal operation, periodicity, symmetry, examples: exponential, sinusoidal, impulse, step functions
- Basics of discrete- and continuous-time systems: interconnection, system properties: memoryless, invertibility, causality, stability, time-invariance, linearity
- Discrete-time and continuous-time linear time-invariant systems: convolution and its properties, impulse response, step response, system stability, block diagram

1. Consider the following continuous-time signal:

$$x(t) = \sin\left(\frac{\pi}{4}t\right) + \cos\left(\frac{\pi}{3}t\right).$$

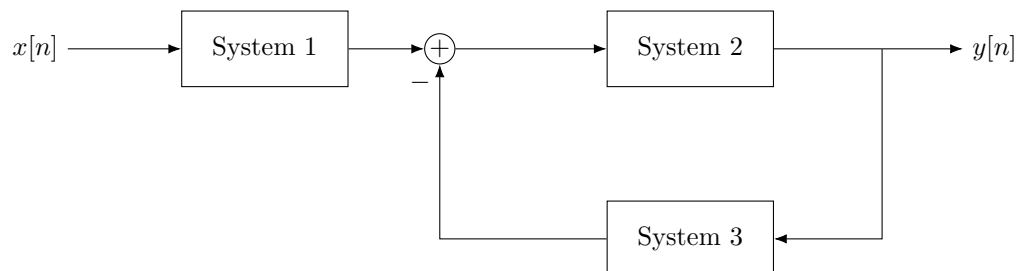
- (i) Compute the energy and average power of $x(t)$;
 - (ii) Determine whether $x(t)$ is periodic and if so, compute its fundamental period;
 - (iii) Repeat (i) and (ii) by changing the index time t to n , i.e., $x[n]$.
2. Consider the following continuous-time systems with input $x(t)$ and output $y(t)$:
 - $y(t) = (x(t-1) + 1)^2$
 - $y(t) = \max(x(t), x(1-t))$

Determine whether they are memoryless, invertible, causal, time-invariant, and linear.

3. Consider the following continuous-time linear time-invariant system that is represented by an impulse response $h(t)$:

$$h(t) = \begin{cases} 2^{-t} & \text{when } 0 \leq t \leq 1 \\ 2^t & \text{when } -1 \leq t \leq 0 \\ 0 & \text{otherwise.} \end{cases}$$

- (i) Determine whether it is memoryless, causal, and stable.
 - (ii) Compute the system's output for the input $x(t) = u(t) - u(t + 1)$.
4. Consider the following interconnected system (w/ input $x[n]$ and output $y[n]$) that consists of three subsystems:
- System 1 with impulse response $h_1[n] = \delta[n] + \delta[n - 1]$;
 - System 2 with impulse response $h_2[n] = \delta[n]$;
 - System 3 with impulse response $h_3[n] = \frac{1}{2}\delta[n - 1]$.



- (i) Find a difference equation that describes the input-output relationship.
- (ii) Assume the condition of initial rest holds. Compute the system's impulse response. Check whether it is stable.
- (iii) Assume that the system is invertible and the condition of initial rest holds for the inverse system, compute the impulse response of the inverse system. Check whether it is stable.